

Days 1-5: Build and Configure a Small Office Network Deliverable: Fully functional LAN with :Proper IP addressing and subnetting. Configured DHCP and DNS services. VLANs for network segmentation. Simulated internet access via a router. Troubleshooting documentation for common issues. Task: Submit a detailed report on the network setup and configurations performed.

Task 1.

Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::20B:BEFF:FE2C:8088
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.10
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time=10ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128

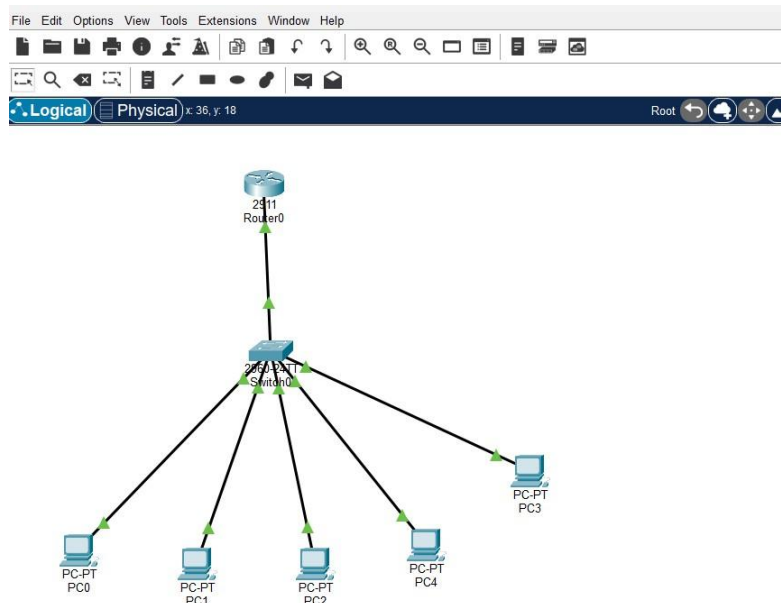
Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.1.13

Pinging 192.168.1.13 with 32 bytes of data:
```

Task 2.

Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.



```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#ip dhcp pool LAN_POOL
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.1.1
```

```

Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip dhcp binding
IP address      Client-ID/      Lease expiration    Type
                Hardware address
192.168.1.2     000B.BE2C.8088  --                  Automatic
Router#show ip dhcp binding
IP address      Client-ID/      Lease expiration    Type
                Hardware address
192.168.1.2     000B.BE2C.8088  --                  Automatic
192.168.1.3     0090.219D.A093  --                  Automatic
192.168.1.4     0003.E419.5B64  --                  Automatic
192.168.1.5     0030.A37A.5402  --                  Automatic
192.168.1.6     0090.2BAD.B8A2  --                  Automatic

```

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20B:BEFF:FE2C:8088

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Task 3.

Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

Steps Performed

1. Opened Notepad as Administrator.
2. Opened the hosts file from:
C:\Windows\System32\drivers\etc\hosts

3. Added the following entry: 192.168.1.1 musicplayer.local
4. Saved the hosts file.
5. Cleared the DNS cache
using: ipconfig /flushdns
6. Opened a web browser and entered: http://musicplayer.local
7. Verified that the router's local web interface opened successfully.

Command Used

```
ipconfig /flushdns
```

Hosts File Entry

```
192.168.1.1      musicplayer.local
```

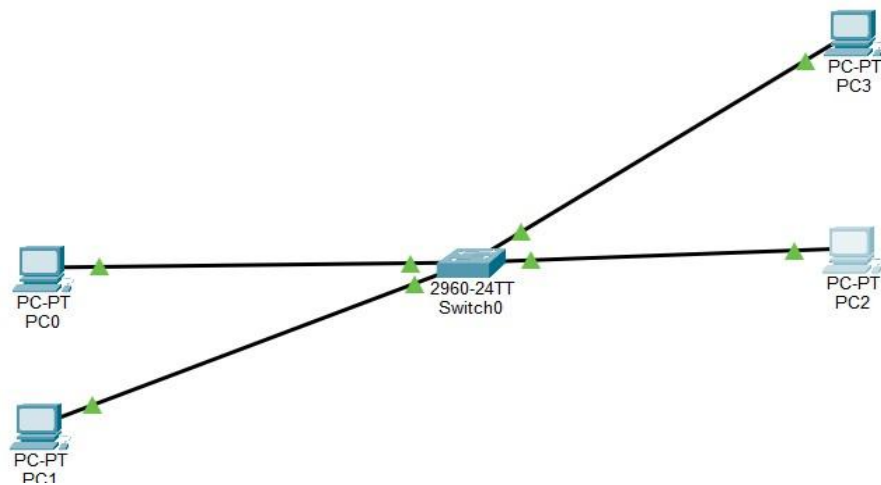
Task 4.

Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.

Hint: Use ping or file sharing to test connectivity between devices in different VLANs.

Configuration Steps

1. Added one switch and four PCs in Cisco Packet Tracer.
2. Created VLAN 10 named Workstations.
3. Created VLAN 20 named Smart Devices.
4. Assigned PC0 and PC1 to VLAN 10.
5. Assigned PC2 and PC3 to VLAN 20.
6. Configured IP addresses for all PCs.
7. Tested connectivity using the ping command.



```

Switch>
Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)#name Workstations
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 20
Switch(config-vlan)#name Smart_Devices
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch#
Switch#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]

```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Task 5.

Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

Configuration Steps

1. Created a LAN consisting of one switch, one router, two PCs, and one external server.
2. Assigned IP addresses to all devices.
3. Configured the router's LAN interface as the NAT Inside interface.
4. Configured the router's WAN interface as the NAT Outside interface.

5. Created an access list to identify the internal network.
6. Enabled NAT overload (PAT) on the router.
7. Configured the PCs with the correct default gateway.
8. Tested connectivity by pinging the external server.